

nication in electronics like time, frequency, phase multiplexing require switching. Switches offer control mechanisms and their use in any kind of an electronic system cannot be understated.

Telescope

The mystery of what lies beyond our vision will forever haunt us. But the telescope is the invention that has helped put all of it in perspective – our place and purpose as a species in the universe, the size of our world, and the space around it. Considered one of the central instruments of the Scientific Revolution of the 17th century, the telescope has brought within our grasp, the solar system and everything beyond the barriers of the earth, proving that the earth is nothing but a big circular rock. It has radically changed who we, as a species, think we are. It has helped give birth to the modern science of astronomy, and hence helped prove/disprove various theories related to earth and the stars.

Credited for the invention of the telescope is Dutch eyeglass maker, Hans Lippershey, who applied for its patent first. However, Galileo Galilei was the first to use it to observe the heavens, and made great improvements in its technology, later finding craters on the moon, and discovering the four largest moons of Jupiter. The first practical telescope though, is credited to Newton since previous telescope designs were rather large and not so easy to use. Over the years, the ability of telescopes to gaze deeper into space has grown exponentially. The famous Hubble telescope (named after Edwin Hubble) orbits the earth once every 96 minutes, and has led to discoveries such as the age of the universe, or the fact that the expansion of the universe is speeding up, contrary to popular belief. The telescope keeps unravelling the mys-

